

1. Dataset Characteristics

The dataset used in this project is a univariate time-series dataset containing historical Brent crude oil prices collected from the U.S. Energy Information Administration (EIA). It is used to forecast future oil price movements based on historical trends and patterns.

The dataset contains 9,011 rows and 2 columns: Date and Price. The Date column serves as the time index, while the Price column represents Brent crude oil prices in U.S. dollars per barrel and is considered the target variable for forecasting.

The data covers the period from 20-May-1987 to 14-Nov-2022, providing approximately 36 years of historical records. The observations are recorded mainly on business days, with gaps during weekends and market holidays.

The dataset is numerical and chronological in nature, making it suitable for time-series analysis and forecasting techniques.

2. Data Cleaning

- **Checked for missing values:** The dataset was inspected for null/missing values using missing-value checking functions.
- **Handled missing values:** Since the file shows the dataset had no significant missing values after checking, no major imputation method was applied.
- **Checked duplicate records:** No duplicate values were found.
- **Outliers Detection:** Outliers were checked using **Z-score analysis** on the **price** column. Since the **Z-score values** were between -3 and 3, the values were not considered outliers, so they were kept in the dataset.
- **Checked missing timestamps:** Missing timestamps were checked using the `pd.date_range()` function by comparing the complete date sequence with the existing timestamps in the dataset.
- **Handled missing timestamps:** Missing timestamps were found and handled using forward filling techniques to maintain continuity in the dataset.

3. Data Transformation

- **Datetime transformation:** The Date column was transformed from text format into datetime format to support time-series processing.
- **Index transformation:** The date column was set as the DataFrame index, converting the dataset into a proper time-series structure.
- **Feature engineering from date:** New time-based features were created from the date column, including:
 - **day of the week**
 - **day of the month**
- **Resampling:** A separate weekly DataFrame was created using resampling to aggregate the daily Brent oil prices into weekly average values for smoother trend analysis.
- **Stationarity transformation:** Differencing was applied to make the time series stationary.
- **Rolling statistics generation:** Rolling mean and rolling standard deviation were created to analyze trends and fluctuations over time.
- **Model-specific transformation for Prophet:** The dataset columns were renamed into Prophet's required format:
 - **Date: ds**
 - **Price: y**
- **Train-test split transformation:** The dataset was split into training and testing subsets based on chronological order, preserving the time sequence.

4. Model Development

The Brent Oil Prices dataset was developed and compared four time-series forecasting models: **ARIMA**, **SARIMA**, **Prophet**, and **Exponential Smoothing (ETS)**.

- **ARIMA Model:** An **ARIMA(1,1,0)** model with a trend component was built using a train-test split. The model was evaluated using **RMSE**, **MAE**, and **MAPE**, and its forecast was compared visually with the actual test values. Time-based cross-validation was also applied using TimeSeriesSplit to measure stability across multiple folds.
- **SARIMA Model:** A **SARIMA** model was developed with seasonal settings to capture possible repeating patterns in the data. The chosen configuration was **order=(2,1,1)** and **seasonal_order=(1,1,1,7)**. Like ARIMA, it was evaluated using a train-test split and then validated further using time-series cross-validation.
- **Prophet Model:** The **Prophet** model was developed as the main forecasting model. It included:
 - **changepoint detection**
 - **seasonality tuning**
 - **external regressors** for major shocks such as:
 - financial crisis
 - COVID-19
 - Russia–Ukraine war
 - oil supply shock

The Prophet model was tested using train-test split, visualized, and then validated using **time-based cross-validation**. After parameter tuning, it showed the best overall balance between accuracy and stability.

- **ETS Model:** An **Exponential Smoothing (ETS)** model was also developed to compare performance with the other forecasting methods. It used trend handling, damped trend, log transformation, and variance stabilization. The model was evaluated using both train-test split and cross-validation.

Model Evaluation and Selection

All four models were compared using:

- **RMSE**
- **MAE**
- **MAPE**

A cross-validation comparison table was created to summarize average performance across folds. Based on the results and visual forecasting behavior, the **Prophet model** was selected as the final forecasting model because it handled trend shifts, structural breaks, and external shocks more effectively than **ARIMA**, **SARIMA**, and **ETS**.

5. Model Deployment

The forecasting model was deployed using the **Streamlit** framework to build an interactive web application for Brent oil price forecasting. The trained Prophet model was saved using the **Joblib** library (joblib.dump) and later loaded into the **Streamlit** application (joblib.load) for real-time forecasting. The deployment included user-friendly features such as forecast horizon selection, historical data visualization, shock event controls, forecast plotting, and downloadable forecast results in CSV format.